

Week 5 — Network Scanning with Nmap

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Scans Performed

- Basic Port Scan: nmap localhost
- Service Version Scan: nmap -sV localhost
- OS Detection Scan: sudo nmap -O localhost

1. Basic Port Scan

```
(kali㉿kali)-[~]
$ nmap localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-15 04:54 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

2. Service Version Scan

```
(kali㉿kali)-[~]
$ nmap -sV localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-15 05:09 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Debian 5 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
```

3. OS Detection Scan

```
(kali㉿kali)-[~]
$ sudo nmap -O localhost
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-15 05:09 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000089s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds
```

What is an Open Port?

- A computer has thousands of "ports" through which it communicates over a network. When a service or program is running and listening for connections, that port is considered "open."
- If no service is running on a port, it stays "closed" and does not respond to connection attempts.

Why Are Too Many Open Ports Risky?

- Every open port is a possible entry point for an attacker. The more open ports a system has, the more ways there are for someone to try to break in.
- Open ports often reveal the exact service and version running on them (as seen in the version scan). If that version has a known vulnerability, an attacker can target it directly.
- Running unnecessary services increases risk without providing any real benefit, since each one adds another exposed point.
- Once an attacker knows the operating system and services running, they can plan a more targeted and effective attack.

Ports Found Open on My Machine

- Only one port was found open: Port 22 (SSH).
- The service version scan identified it as running OpenSSH 10.2p1 (Debian), protocol 2.0.
- The OS detection scan identified the machine as running Linux (kernel range 2.6.32 - 6.2), classified as a general-purpose device.
- Having only one open port is a good security practice, since it keeps the attack surface small. However, this single port (SSH) is still exposed and could be targeted through brute force attempts or if the SSH version has a known vulnerability, similar to the Brute Force attack practiced earlier in this internship.